

Deep Learning

What is Deep Learning



Prof. Anxiao (Andrew) Jiang

Learning Objectives:

1. Understand what deep learning is.
2. Understand what a neuron is.
3. Understand how to train a neural network.



Roadmap of this lecture:

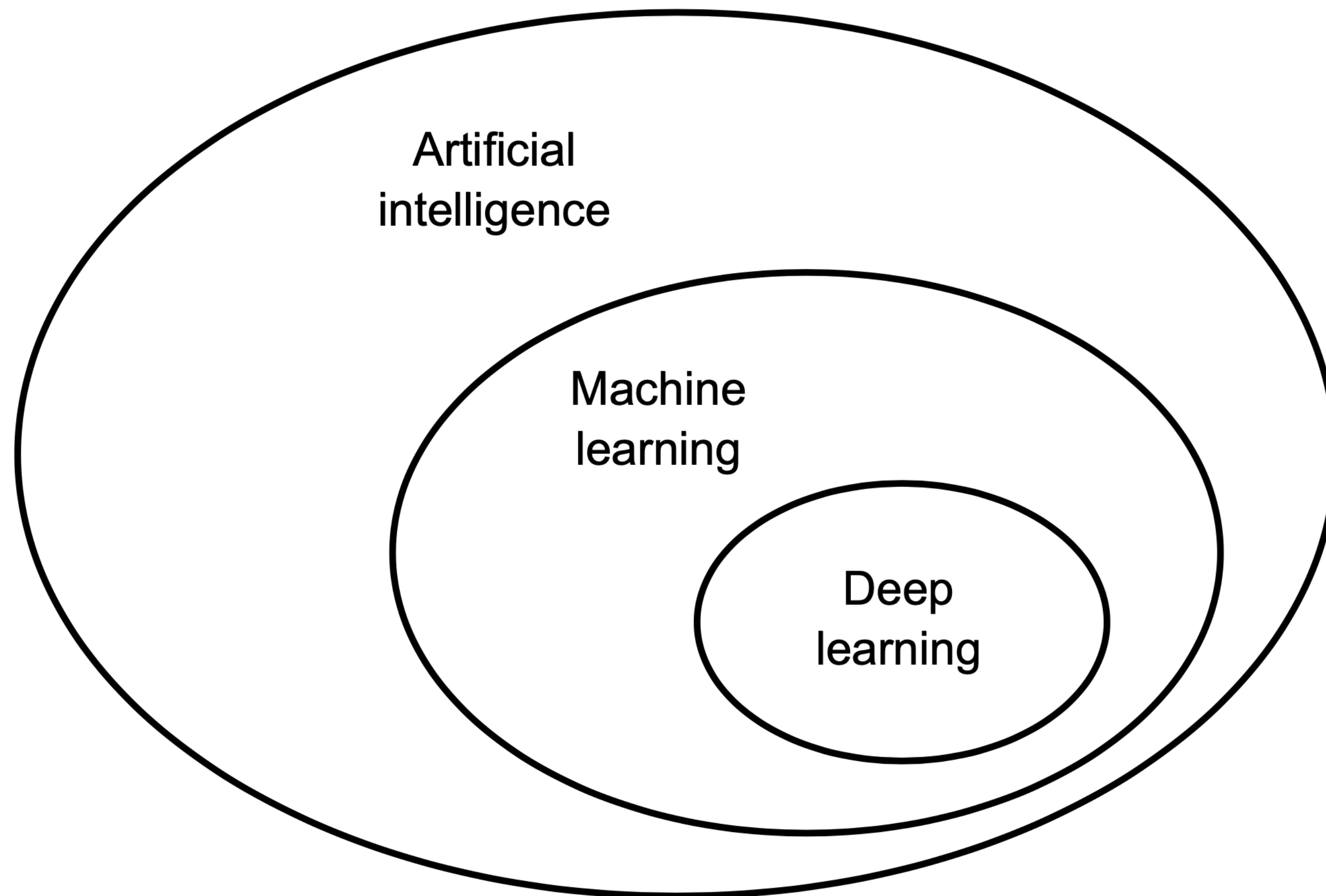
1. What is machine learning and deep learning.

2. What is feature extraction.

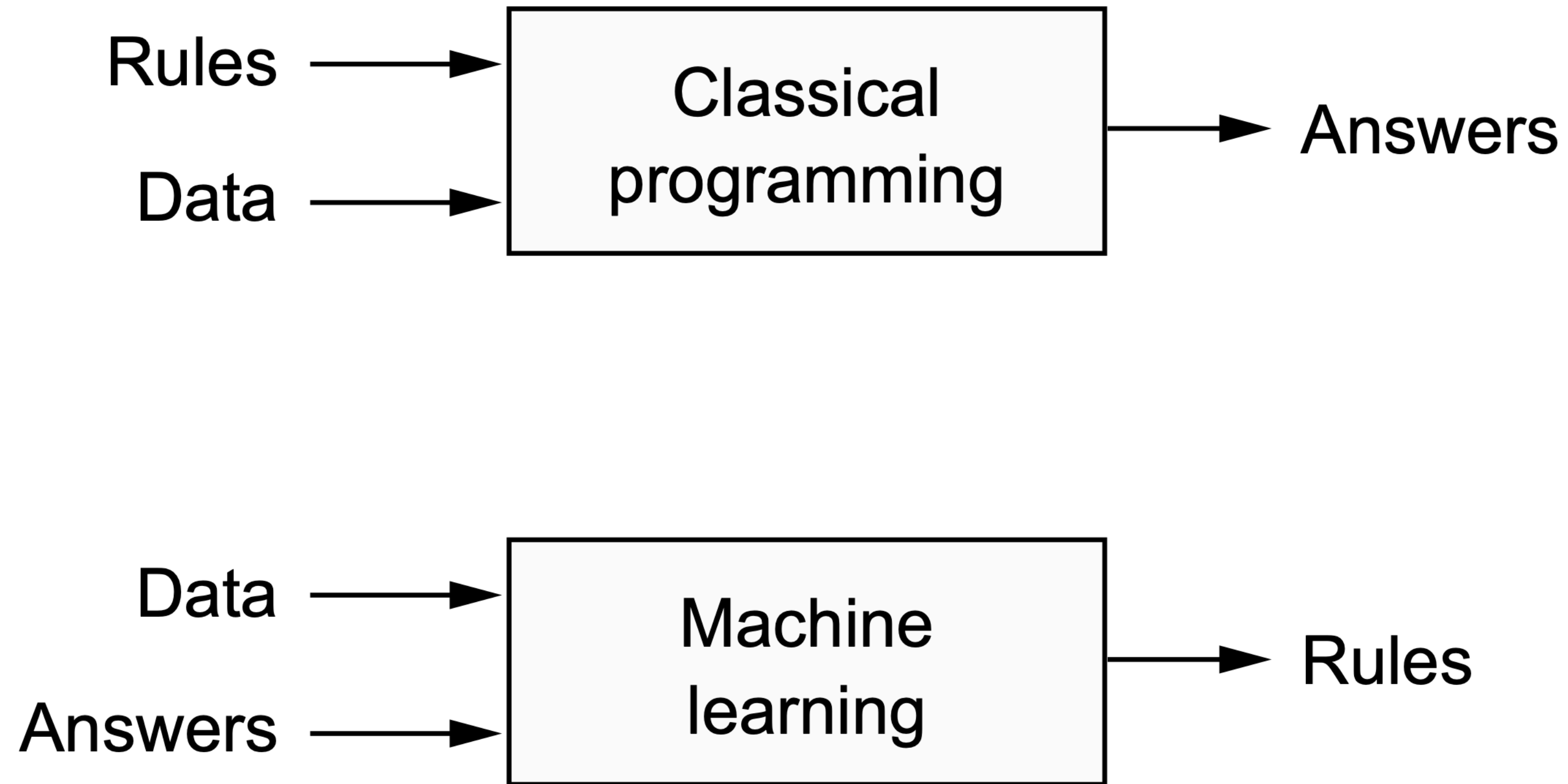
3. What is a neuron and a neural network.

4. How to train a neural network.

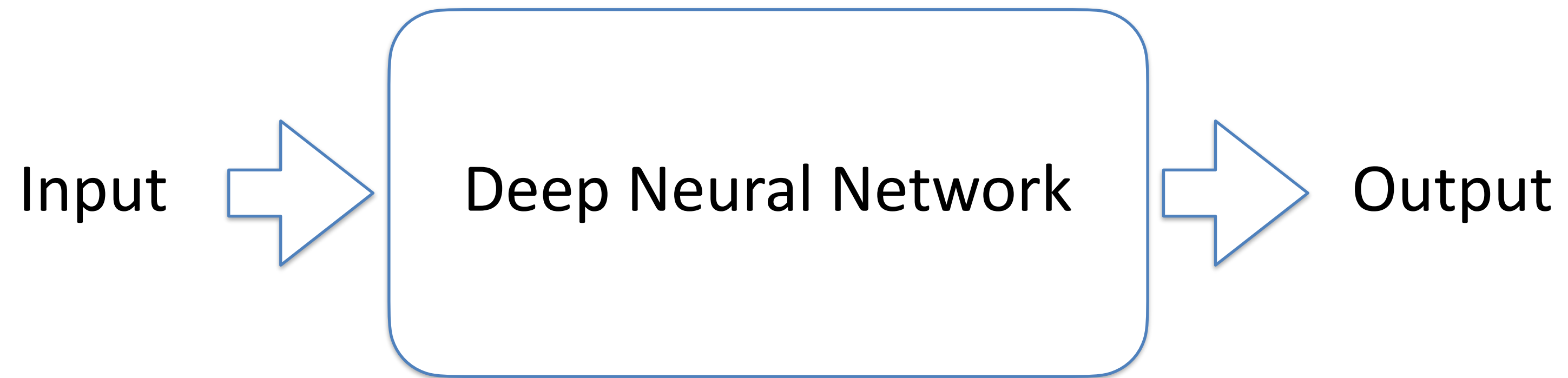




Classical algorithms vs. **Machine Learning**



Deep Learning and **Neural Network**



Quiz questions:

1. What is the difference between machine learning and algorithm design?
2. What is the relation between machine learning and deep learning?

Roadmap of this lecture:

1. What is machine learning and deep learning.

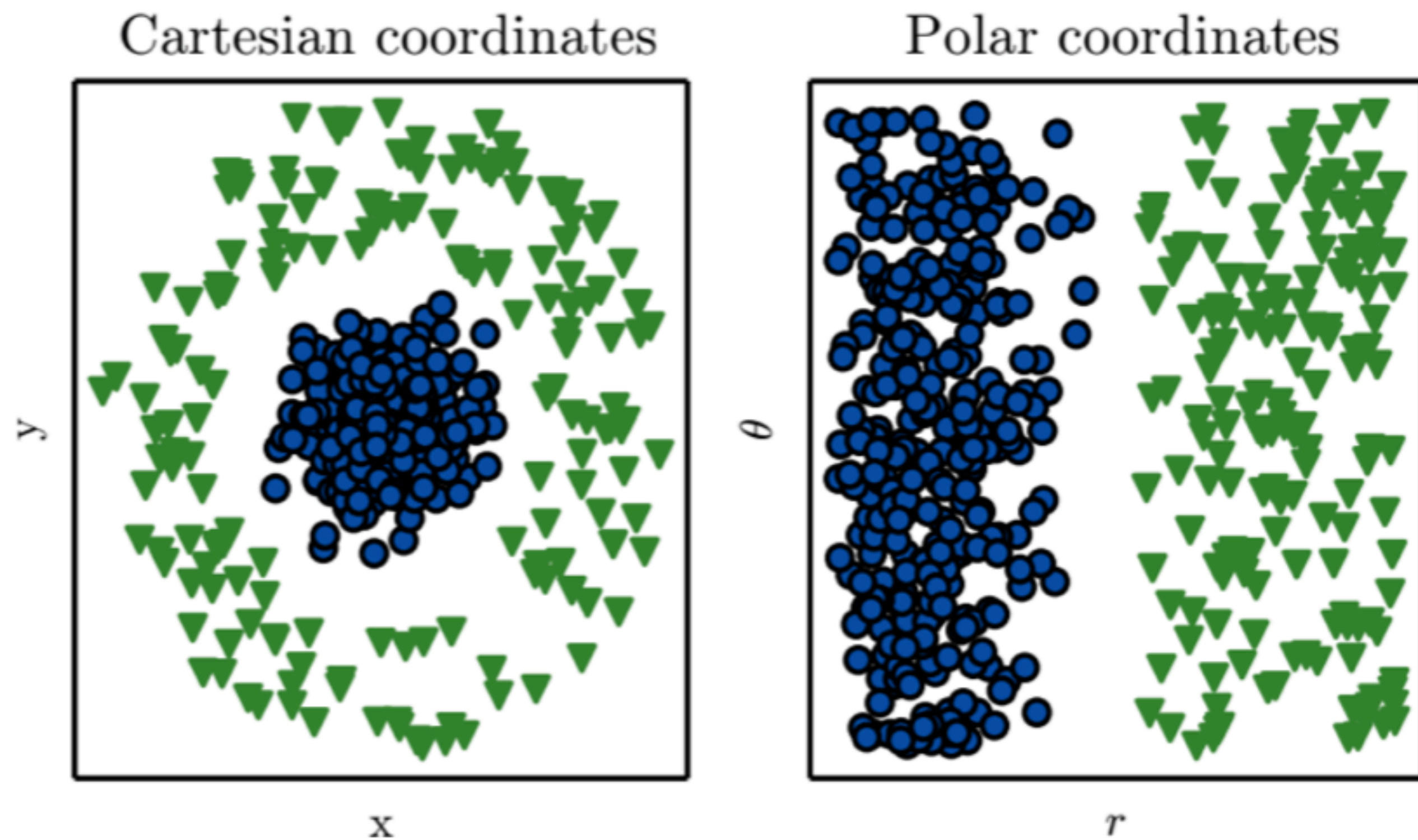
2. What is feature extraction.

3. What is a neuron and a neural network.

4. How to train a neural network.



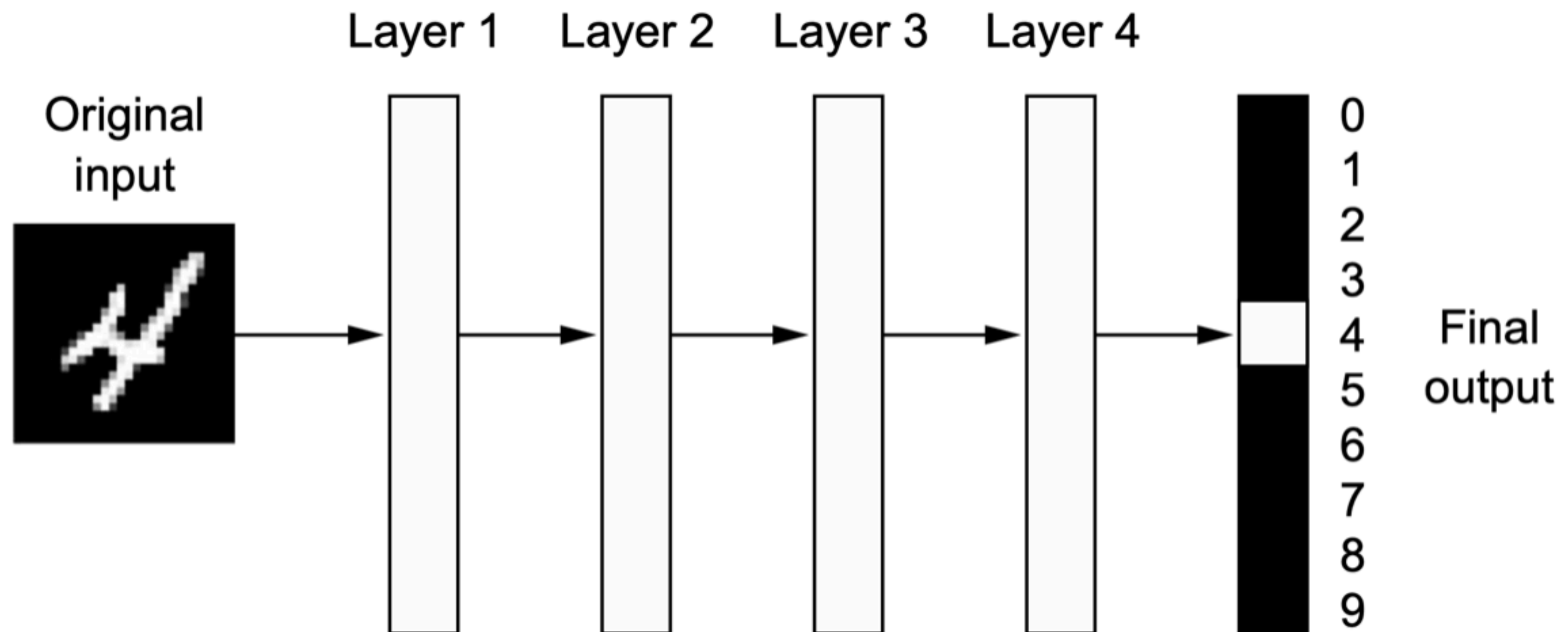
Example of Transformation



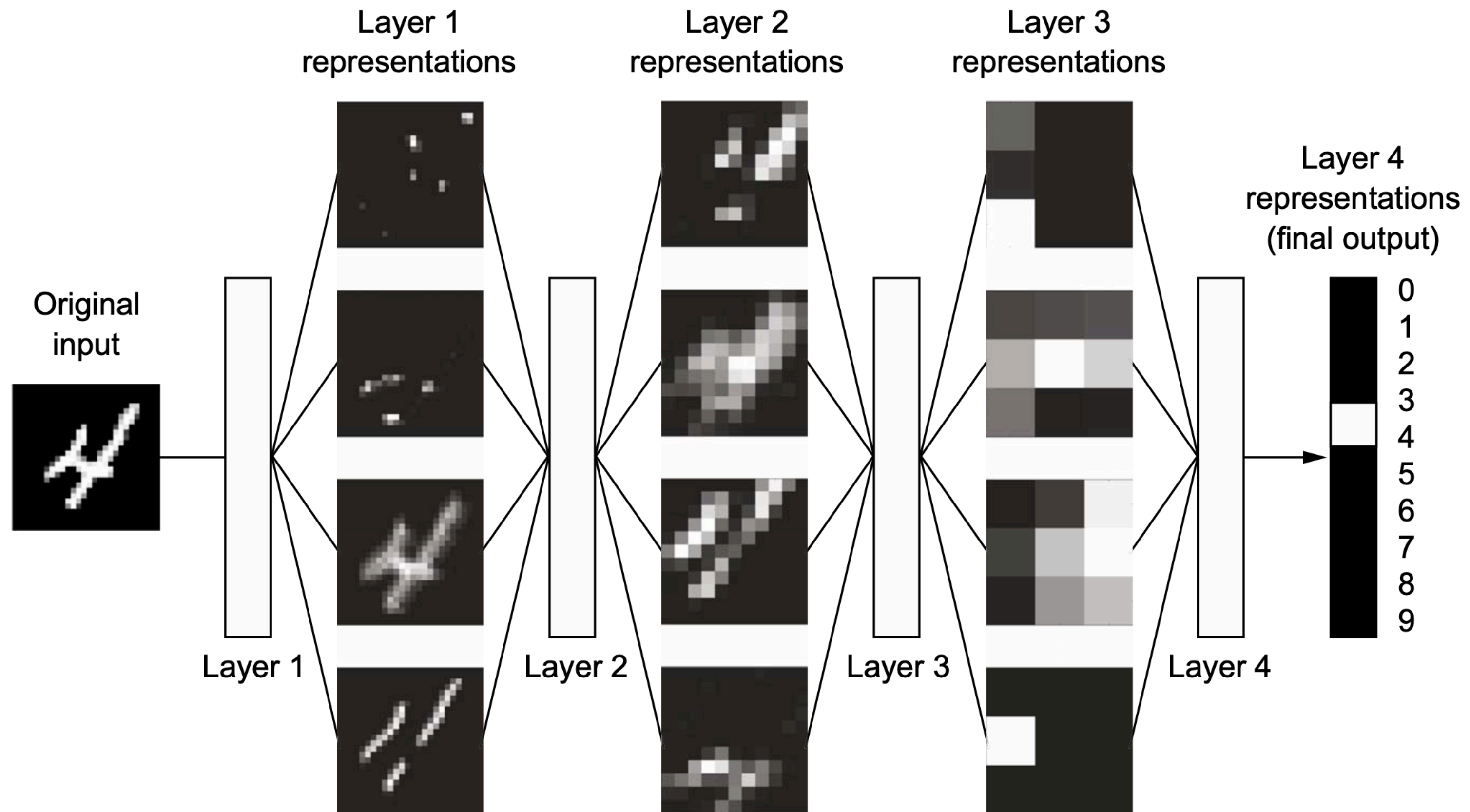
Example of Transformation



Example: handwritten digit recognition



What features are detected in different layers



Quiz question:

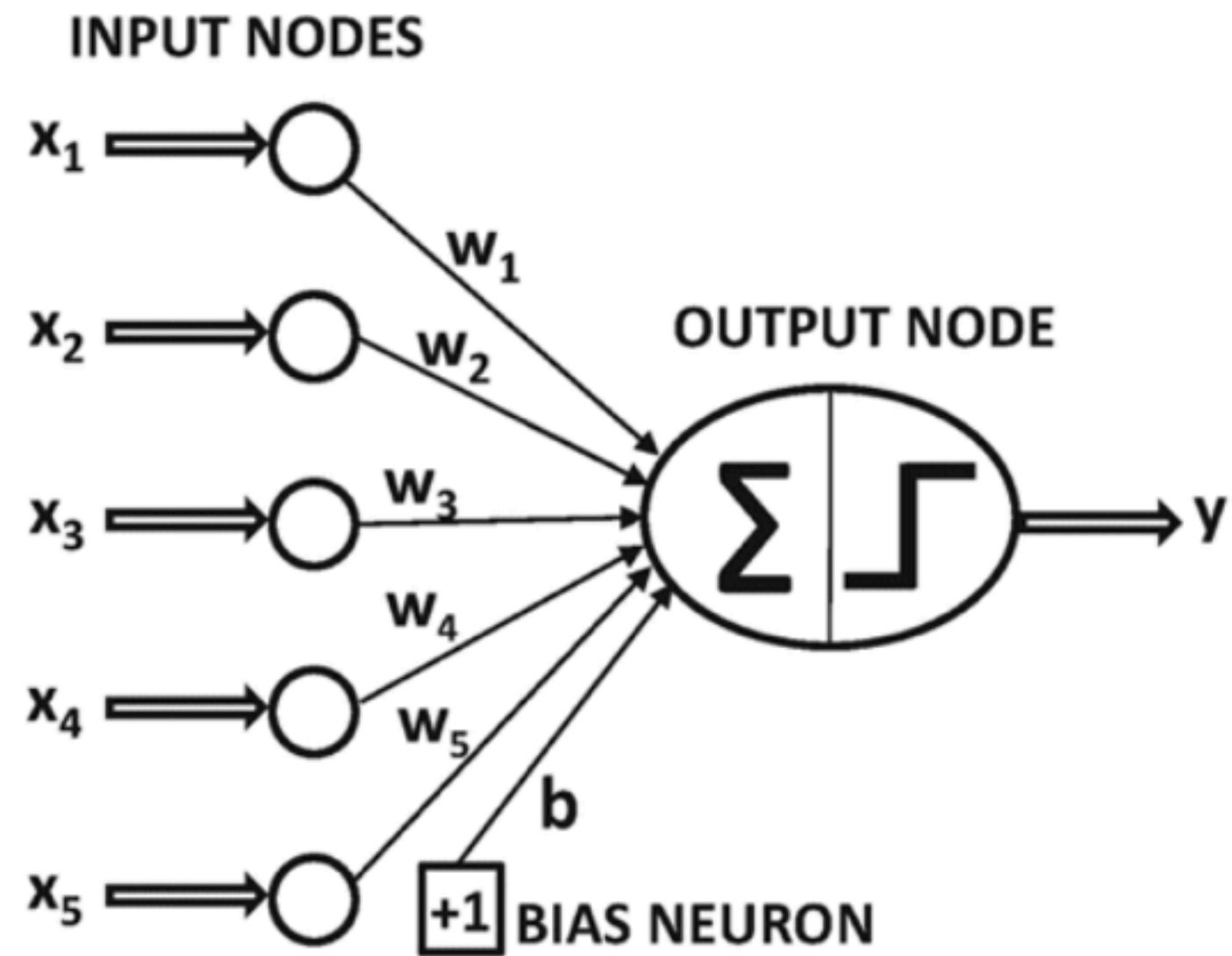
I. What is the importance of feature extraction?

Roadmap of this lecture:

1. What is machine learning and deep learning.
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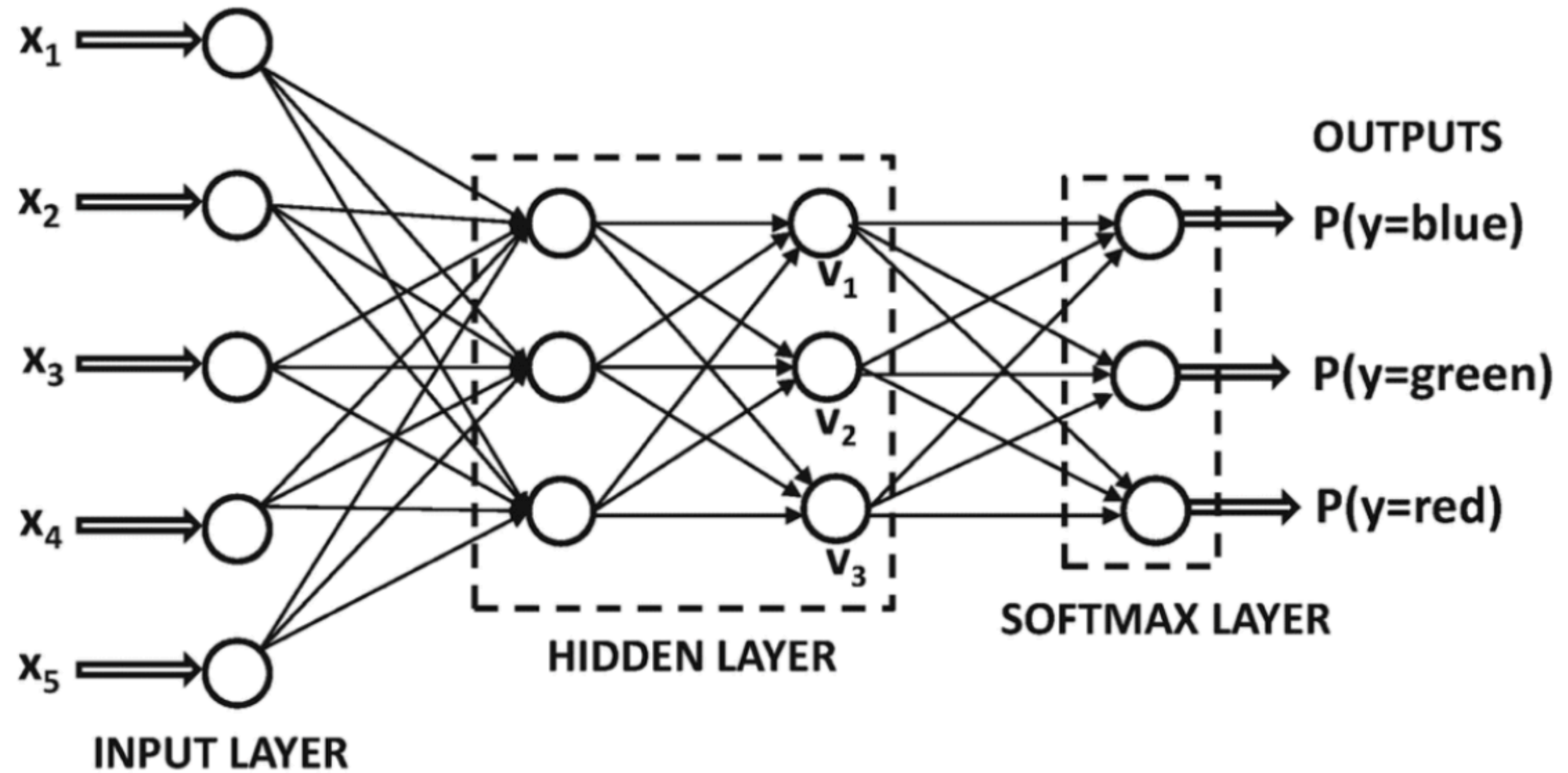


Neuron



$$y = \text{sign}\{\bar{W} \cdot \bar{X} + b\} = \text{sign}\left\{\sum_{j=1}^d w_j x_j + b\right\}$$

Neural Network



Quiz questions:

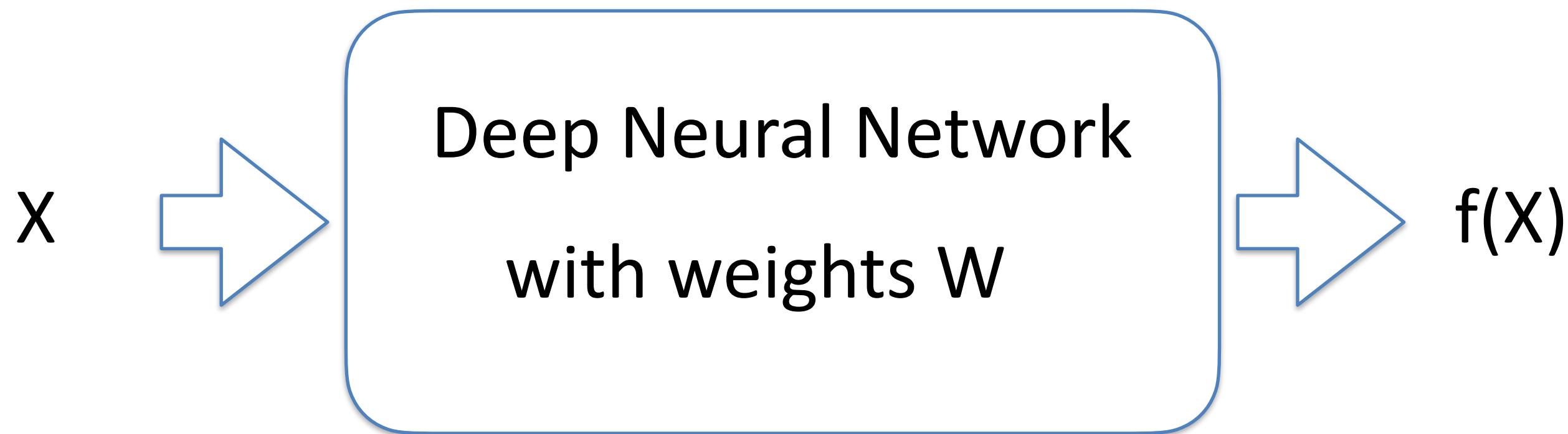
1. What is a neuron?
2. What is a neural network?

Roadmap of this lecture:

1. What is machine learning and deep learning.
2. What is feature extraction.
3. What is a neuron and a neural network.
4. How to train a neural network.



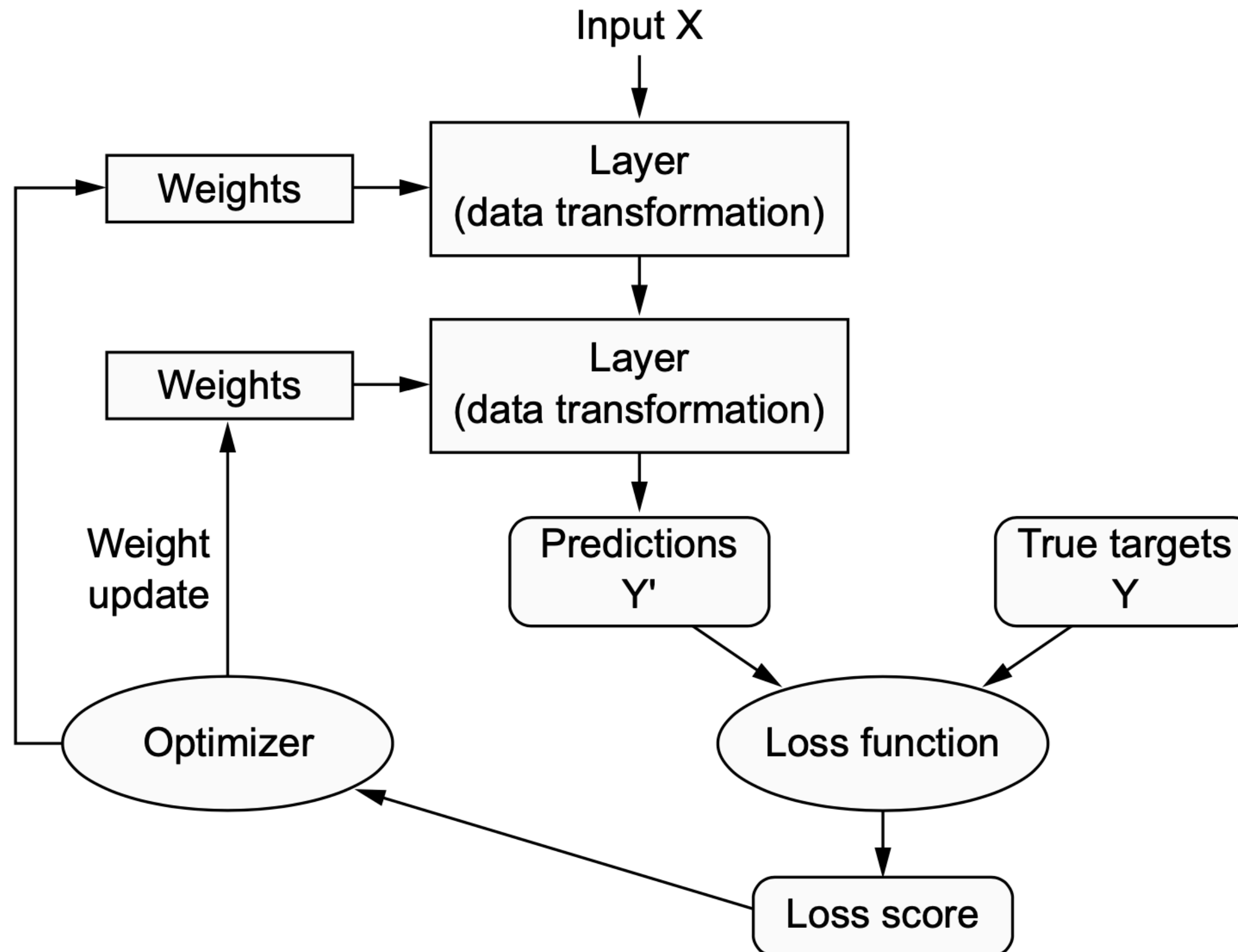
Train Neural Network



Need: a dataset, where every sample is $(X, f(X))$

Task: adjust weights W , to minimize the difference between the target output and the model's real output

Train Neural Network



Applications of Deep Learning

- Computer vision (smart camera, robot, self-driving cars, etc.)
- Natural language processing (machine translation, chatbot, etc.)
- Game playing (alpha Go, video games, etc.)
- Create art or products (painting, music, poem, fashion, etc.)
- Data storage and transmission (data compression, transmission, etc.)
- Finance and economy (trading, recommendation, economy survey, etc.)
- Healthcare (read X-ray pictures, diagnosis, drug design, etc.)
- Physics, business, education, smart homes, etc. (More and more applications every day.)

Quiz questions:

1. What is the process of training a neural network?
2. Why do we need lots of data for training a neural network?